

THE KCP LIMITED

FIRE PROTECTION

Fire Protection

- Protect employees, property, and facilitate business continuity
- 4 objectives of fire protection programs
 - Preventing fires
 - Detecting and responding to fires
 - Detect fires early
 - Initiate appropriate alarms
 - Respond quickly to alarms
 - Controlling, suppressing, and extinguishing fires
 - Recovering from fires



Preventing Fires

- Non-combustible construction materials
- Configuring appropriate fire area separations
- Uses of the building
- Overlap to other objectives

Detecting and Responding to Fires

- Human observers
- Electrical and mechanical devices for detection
- Activate an alarm and sound
- Response of occupants in building

Controlling, Suppressing and Extinguishing fires

- Prevention activities have failed and fire starts, program for control, suppression, and extinguishing
- Control – physical barriers to contain fires and products of combustion
- Barriers – fire-rated walls, doors, windows, and air-handling dampers
- Suppression – automatic sprinkler systems
- Human – fire extinguisher use, brigades, local fire departments

Recovering from Fires

- Secure the scene
- Begin investigation
- Resume normal operations if possible

Fire Hazard Analysis

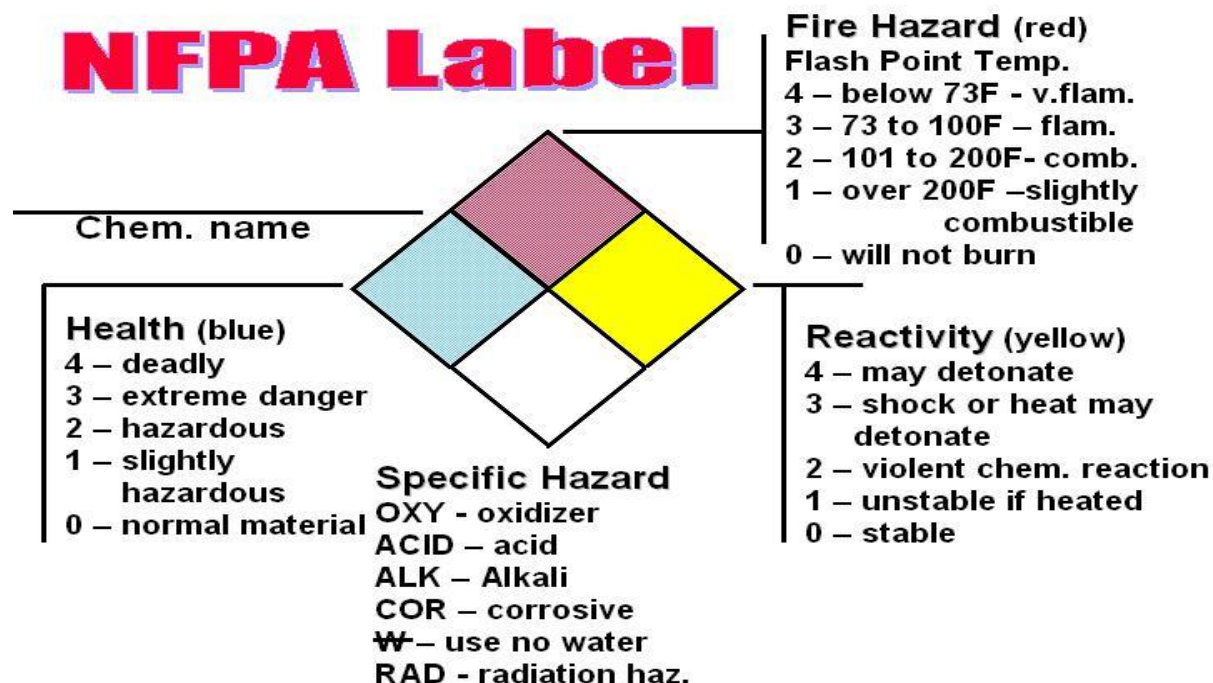
- Fire hazards need to be assessed before, during, and after construction of a facility.
- Topics that should be analyzed:
 - site
 - building construction
 - building content
 - management factors
 - people factors (numbers and characteristics)
 - fire protection system (if for an existing facility)
 - after the fire

Identification of Hazardous Materials

- Parameters: degrees of toxicity, flammability, and instability
- Safety Data Sheets (SDS) and NFPA 704
- NFPA Hazardous material identification system
 - Diamond-shaped symbol with colored numerals and backgrounds
 - Categories: health hazard (blue), flammability hazard (red) and instability hazard (yellow)
 - Numeral ratings: 0–4 higher numbers higher degree of hazard

NFPA Identification System

- The NFPA Identification System is a quick way to identify hazardous properties of chemicals.
- Hazards rated 0–4, with 0 being no danger and 4 being extreme or high danger:



NFPA Hazards

- Health (BLUE)
 - This hazard can gain exposure from contact, inhalation, or indigestion.
 - It can cause injury from direct or indirect exposure.
- Fire or flammability (RED)
 - This hazard alerts someone of the flash point of the chemical.
- Reactivity (YELLOW)
 - This hazard signals if the chemical will have a chemical reaction to heat or vibrations.

Shipping Regulations

- Shipping of hazardous materials are regulated by the U.S. Government's Department of Transportation
 - They are also regulated by 49 CFR, Parts 170–180.

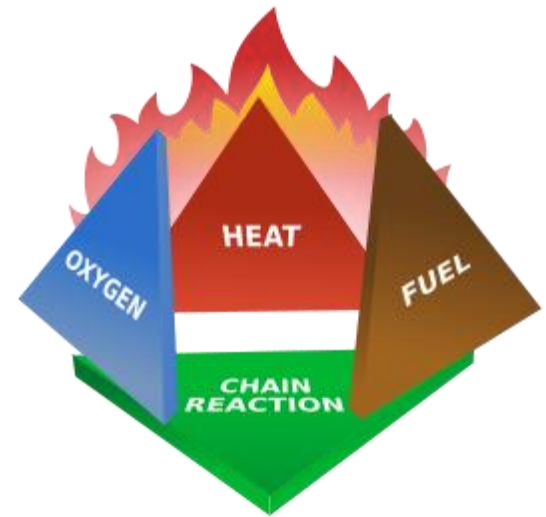


Evaluating Fire Hazards

- When changing any aspect of production, whether it be a method or a new facility, there are several questions that should be asked:
 - What materials are flammable?
 - Where are flammable and combustible materials located?
 - What toxic gases might evolve in a fire?
 - How much time might it take for a fire to spread to other areas?
 - How many people are likely to be involved in the facility?

The Chemistry of Fire

- Process of combustions
 - fuel, oxygen, heat and chemical chain reaction
 - heat energy released in self-catalyzed reaction of condensed-phase fuel or gas-phase or both
 - rapid oxidation of fuel by oxygen in air
 - confined process – explosion
 - combustion process emit heat and light



Controlling Fires

- Cool a fire
 - Lowering the heat of the fire can reduce the size of the fire.
- Remove fuel from a fire
 - A fire will extinguish with no fuel supply. This can be difficult and dangerous.
- Limit oxygen in a fire
 - Limiting the oxygen in a fire can be done by smothering the area with a noncombustible material or throwing sand or dirt on the fire.
- Interrupt the chain reaction in a fire
 - Using a fire extinguisher will limit the oxygen to the fire and will interrupt the chain reaction in a fire by removing the free radicals, which are the lifeblood of the fire.
- Use extinguishing agents
 - Extinguishing Agents Can Sometimes Attack More than One of the Four Component of the Tetrahedron, This Could Result in Eliminating the Fire Quicker

Classification of Fires

- Class A Fires

(usually occur in ordinary materials, like wood, paper, rags, & rubbish)



- Class B Fires

(usually occur with a vapor-air mixture over the surface of flammable liquids such as gasoline, oil, grease, and paints)



Classification of Fires (Cont.)

- Class C Fires
(electrical fires)



- Class D Fires
(usually occur in combustible metals such as magnesium, titanium, and potassium)



Classification of Fires (Cont.)

- Class K Fires
(usually involve cooking greases
or cooking oils)



Fire Prevention: Construction of Facilities

1. Objectives of fire prevention
2. Planning for fire protection
3. Site planning
4. Construction materials and interior furnishings
5. Fire protection methods and concepts in building design

1. Objectives of Fire Prevention

- In a building fire, first protect life and property second.
- Design and construction must account for a wide range of fire safety features.
- Interiors and contents of a building must be protected from dangers of fires, but there must be adequate water supplies and easy access for the fire department.
- National, state, and local codes provide for minimal measures for fire safety.
- Planning and construction based on such codes should not reduce or limit fire-safe design efforts.

2. Planning for Fire Protection

- Continuity of operations
 - The designer needs to think about how long the downtime would be in the event of a fire.
- Property protection
 - Which parts of the facility need to be more protected because they are considered high-risk for fire and/or the productivity of the operations risk is too high.
- Concerns
 - Life safety: Who will use the building? What will the people using the building be doing most of the time?

2. Planning for Fire Protection (Cont.)

- Fire hazards in buildings
 - In the event of a fire, what, how, and who would be affected by the fire?
 - Heat and flames
 - 34% of deaths were a result for the actual heat and flames of the fire.
 - Smoke and gas
 - 66% of deaths were a result from smoke and toxic gases that were produced as the fire evolved.

3. Site Planning

- Items to consider when planning fire safety for sites:
 - Traffic and transportation
 - Can fire vehicles respond to the fire during heavy traffic?
 - How long will it take a fire vehicle to reach the facility?
 - Fire department access to the site
 - Can the fire vehicles reach all the building on site?
 - Fire department access to facility's interior
 - Can the fire vehicles gain access to the fire area?
 - Water supply to the site
 - Is there enough water available to extinguish a fire for the whole facility?

4. Exposure Protection

- Space between building needs to be enough so that the fire doesn't spread from one building to another because of the location

Construction Materials and Interior Furnishings

1. Heavy timber construction
2. Noncombustible and limited-combustible construction
3. Ordinary construction
4. Wood-frame construction
5. Interior finish

1. Heavy-Timber Construction

- Heavy-timber construction is characterized by masonry walls, heavy-timber columns and beams, and heavy plank floors.

2. Noncombustible and Limited Combustible Construction

- Exposed steel beams and columns, masonry, metal, and gypsum wallboard are the most common types of noncombustible and limited combustible construction, which are not fire resistant

3. Ordinary Construction

- Consists of masonry exterior-bearing walls, or bearing portions of exterior walls that are noncombustible
- To prevent the free passage of flames through concealed spaces or opening:
 - Trim all combustible framing away from sources of heat
 - Provide effective fire barriers against the spread of fire between all subdivisions and all stories of the building
 - Provide adequate fire separation against exterior exposure
 - Fire-stop all vertical and horizontal draft openings to form effective barriers to stop or slow the spread of fire

4. Wood-Frame Construction

- Wood-frame construction consists primarily of wooden exterior walls, partitions, floors, and roofs

5. Interior Finish

- Types of Interior Finish
 - plastics
 - wood
 - steel
 - concrete
 - glass
 - gypsum
 - masonry

Fire Protection Methods and Concepts in Building Design

1. Confining fire
2. Controlling smoke
3. Exits
4. Evacuation
5. Ventilation
6. Fire doors
7. Connections for sprinklers and standpipes

1. Confining Fire

- Confining fires can be done during the designing process, with stair enclosures and fire walls, separate units, and fire doors



2. Controlling Smoke

- Controlling smoke can be done by regulating the ceiling heights and ventilation when constructing and operating the building.

3. Exits

- Exits should be lit and clearly mark, there should be ample amounts of exits so that evacuation is fast for occupants.

4. Evacuation

- Evacuation should be practiced so that in the event of a real emergency everyone knows their role and what to do.

5. Ventilation

- Ventilation is vital in removing smoke, toxic gases, and heat; it requires that appropriate skylights, roof hatches, emergency escape exits, and similar devices be installed. It allows release of unburned combustible gases and prevents spread of fire to unburned areas.

6. Fire Doors

- Fire doors provide protection of horizontal openings. Doors are rated from $\frac{3}{4}$ –3 hours. Doors must be closed in a fire to have any value. Maintain good housekeeping and inspect routinely.

7. Sprinklers & Standpipes

- Connections must be carefully located and clearly marked. Size of building determines the volume and pressure of water required for building. Design for use of water is essential for water use.

Fire Protection in the Computer Room

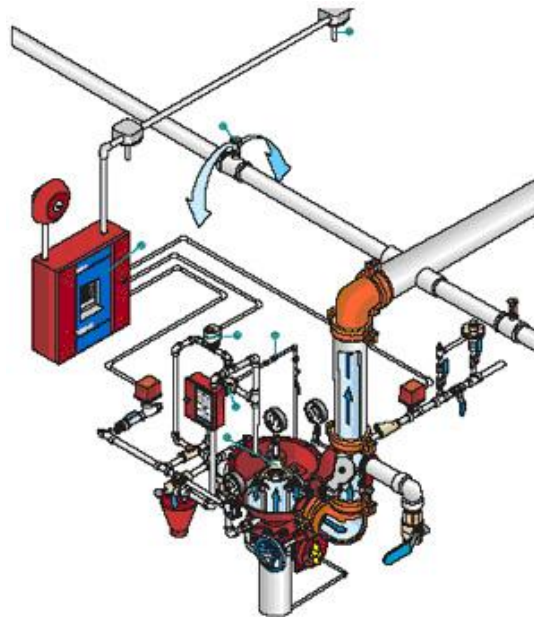
- Fire suppression should keep in mind:
 1. extinguish fires before damage or injury is done
 2. allow workers to escape the area unharmed
 3. protect vulnerable electronic hardware and software

Fire Prevention: Maintaining Facilities

1. Inspections
2. Hot-Work Permits
3. Training Employees
4. Communications

1. Inspections

- Should be conducted for every operation in the facility:
(e.g., control valves, hydrants, fire pumps, hose houses, sprinkler systems, portable fire extinguishers, fire doors and exits, control rooms, alarms and communication equipment)



2. Hot-Work Permits

- Hot-work permits are established to control unwanted fires from sparks or open flames.
- Workers should:
 - inspect the area where hot work is to be done
 - establish a fire watch during and after hot-work is completed for 30 minutes
 - have fire extinguishers on hand
 - communicate with all departments
 - limit the amount of flame or sparks generated
- Safety coordinator needs to establish who is responsible for hot work conducted by outside contractors

3. Training Employees

- Employees should know when to use a portable fire extinguisher and when to evacuate the premises.
- When using a portable fire extinguisher employees, should execute the **P.A.S.S.** Method.



4. Communications

- Communication should be done to let other employees know when there is a fire and also that the fire brigade or fire department will attend to the situation.

Facility Fire Protection Program

- Employees should know their roles in these procedures:
 - Immediately detect the fire and promptly transmit an alarm
 - Initiate evacuation of the building
 - Confine the fire
 - Extinguish the fire

Factors Contributing to Industrial Fires

- Electrical equipment
- Smoking
- Friction
- Foreign objects or tramp metal
- Open flames
- Spontaneous ignition
- Housekeeping
- Explosive atmospheres

Fire Detection

1. Human Observer
2. Automatic Fire-Detection Systems
3. Building Elements and Contents
4. Elements of Building Fire Safety

1. Human Observer

- One of the most reliable detection devices if the observer is paying close attention to his/her surroundings
 - can alert the proper authority at any signs of fire, flames, and/or smoke
 - can act quickly in an emergency and use a portable fire extinguisher
 - can give a vivid report as to why the incident happened and what further precautions need to take place

2. Automatic Fire-Detection Systems

- Thermal detectors
 - Fixed-thermal detectors
 - Rate-compensated thermal detectors
 - Rate-of-rise thermal detectors
 - Line thermal detectors
 - Eutectic-salt-line thermal detectors
- Bulb detection system
- Smoke detectors
- Beam photoelectric detectors
- Reflected-beam photoelectric detectors
- Products-of-combustion (ionization) detectors
 - Single-chamber ionization detectors
 - Dual-chamber ionization detectors
- Low-voltage ionization detectors
- Flame detectors
- Infrared detectors
- Ultraviolet detectors
- Combustion-gas detectors
- Extinguishing-system attachments
- Sensor systems

3. Building Elements and Contents

- Building elements and contents may result in the spread of a fire and in releasing toxic gases.

4. Elements of Building Fire Safety

- The elements of building safety need to be addressed from the start with the design, construction, and good management of the facilities.
- Good housekeeping is one of the major factors in both fire prevention and fire control.

Facility Fire Protection Program

1. OBJECTIVES OF A FIRE PROTECTION PROGRAM

2. FIRE PROTECTION ENGINEERING

3. FIRE DRILLS

4. FIRE BRIGADES

1. Objectives of a Fire Protection Program

- No Facility is Absolutely Fire-Resistive
- Fire and Flame can Spread Horizontally and Vertically
- Heat, Smoke, and Toxic Gases are Possibly the Greatest Danger to Life
- Heat Energy can be Transmitted by
Convection, Conduction, Radiation, Direct Flame Contact

2. *Fire Protection Engineering*

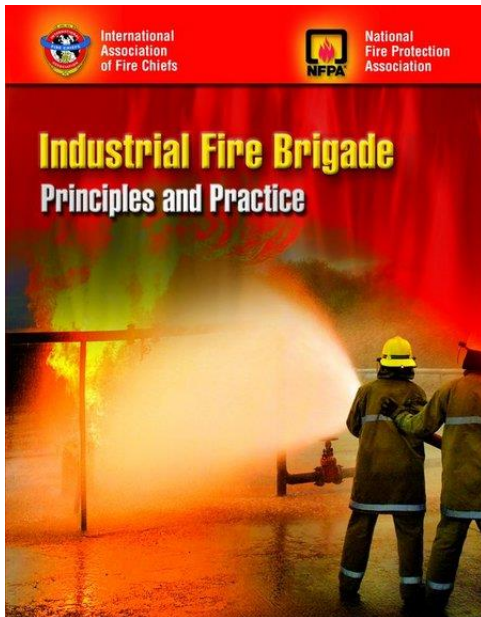
3. *Fire Drills*

- Fire Protection Engineers Work with Others to Achieve the Most Efficient Fire Protection
- Fire Drills Need to be Carefully Planned and Executed Effectively
- Fire Exits and Alternate Routes Need to be Posted and Clearly Marked



4. Fire Brigades

- 29 CFR 1910.156 must be met when forming an industrial fire brigade
- Industries should consult with the municipal fire department to gain knowledge on training and regulations



Alarm Systems

1. Protected Premises (Local) Alarm System
2. Auxiliary Alarm Systems
3. Supervising Station Fire Alarm System
 - Central Station Systems
 - Proprietary Alarm Systems
 - Remote Supervising Station Alarm System
4. System Spacing of Detectors

1. Local Alarm Systems

- Local alarm systems consist of bells, horns, lights, and sirens
- Generally used for life protection to evacuate the occupants and limit injury or loss of life from the fire

2. Auxiliary Alarm Systems

- Auxiliary alarm systems are triggered at the sight of the incident and are tied to protected-premises fire alarm systems.
 - When protected-premises fire system activates, the auxiliary fire alarm system signals the public fire service communications center.

3. Supervising Station Fire Alarm System

- Central station systems—for-hire monitoring companies.
 - These companies are a leased station in which trained personnel may contact the local fire department in the event of an emergency.
- Proprietary alarm systems—operated on behalf of the facility under one owner and continually monitor all aspects of the emergency system related to all of the one owner's facilities.
- Remote supervising alarm systems—limited in scope to the alarm, supervisory, or trouble, systems of one or more specific protected premises fire alarm systems.

4. Spacing of Detectors

- The spacing, location, and maintenance of detectors is a main concern
 - type of building, process, and materials used will determine the system design considerations

Portable Fire Extinguishers

1. Principles of Use
2. Selection of Extinguishers
3. Types of Extinguishers
4. Miscellaneous Equipment
5. Maintenance and Inspection

1. Principles of Use

- Classification of fire extinguishers
 - Ability to handle specific classes and sizes of fires
- Location of fire extinguishers
 - Keep close to hazards so that they are easily accessible , but not so close that they would be damaged by a fire
- Distribution of extinguishers
 - Determine the minimum number and type of portable extinguishers that should be installed on a given floor or area

2. Selection of Extinguishers

- When selecting a portable fire extinguisher, one needs to know which types of fires are possible in the area.
- There are different types of extinguishers for the 5 types of fires
 - Class A, Class B, Class C, Class D, & Class K

3. Types of Portable Extinguishers

- Water-solution extinguishers



- Dry-chemical extinguishers
- There are 4 types:
 - Sodium Bicarbonate
 - Potassium Bicarbonate
 - Potassium Chloride
 - Ammonium Phosphate

3. Types of Portable Extinguishers (Cont.)

- Carbon dioxide extinguishers
 - displace the available oxygen in a fire
- Dry-powder extinguishers:
 - G-1 powdered agent
 - Met-L-X
 - Lith-X
 - Met-L-Kyl

4. Miscellaneous Equipment

- Wheel equipment: when a larger extinguishing agent such as a 75- or 100-lb unit is needed
- Wheeled “twinned” extinguishers: contain both Purple K dry chemical and “light water” fluorocarbon foam; completely protects against re-flash
- Vehicle-mounted equipment: for transportation of large amounts of extinguishing agent
- Fire blankets: small fires, extinguish burning clothes or small, open containers



5. Maintenance and Inspection

- Maintenance of portable fire extinguishers, no matter the size or type, needs to be in accordance with NFPA 10
 - Tags should be placed on extinguishers with the date of inspection.

Sprinkler and Water-Spray Systems

1. Water Supply and Storage
2. Automatic Sprinklers
3. Water-Spray Systems
4. Fire Hydrants
5. Fire Hoses
6. Hose Nozzles

1. Water Supply and Storage

- Water may be supplied from the following:
 - Underground supply mains from public water works
 - Automatically or manually controlled pumps drawing water from lakes, ponds, rivers, surface storage tanks, underground reservoirs, or similar adequate sources
 - Pressure tanks containing water in a quantity determined by the formula in NFPA 13
 - Elevated tanks or reservoirs that depend on gravity to force water through the system

2. Automatic Sprinklers

- Wet-pipe system
 - water is present in the system all the way to the to the sprinkler head
- Dry-pipe system
 - compressed air holds water back in the system keeping the pipes and sprinkler head dry until needed
- Pre-action systems
 - similar to dry-pipe system, but water is released only where there is a fire detected and not throughout the entire system
- Deluge systems
 - activated by fire detectors and supply water to large areas to prevent the spread of fire

3. Water-Spray Systems

- Effective on all types of fires when there is no hazardous chemical reaction between the water and the burning materials
- Water-spray systems serve the following purposes:
 - Extinguish fires
 - Control fire where extinguishing is not effective, such as gas leaks
 - Exposure protections—absorb heat transferred from equipment by the spray
 - Prevent fire by having water spray dissolve, dilute, disperse, or cool flammable materials

4. Fire Hydrants

- Fire hydrants are sometimes needed when a facility is not within reach of a public hydrant.

5. Fire Hoses

- Fire hoses need to be available for immediate use, they need to be easy to reach, and the space around the hoses needs to be free and clear of any debris.

6. Hose Nozzles

- Hose nozzles must be able to supply the firefighter with the correct pressure and amount of water, that's why there are many types of nozzles; nozzles are also available for foam and dry-chemical agents.

Special Systems and Agents

1. Foam and Foam Systems
2. Carbon Dioxide Extinguishing Systems
3. Dry-Chemical Extinguishing Systems
4. Wet Chemical Extinguishing Systems
5. Water Spray and Automatic Sprinkler Systems
6. Preventing Explosions
7. Suppressing Explosions

1. Foam Systems

- Foam is often used to protect dip tanks, oil and paint storage rooms, and asphalt coating tanks
 - low-expansion foam
 - chemical foam
 - mechanical or air-generated foam
 - protein concentrates
 - fluorinated surfactant foams
 - foam-water systems
 - wet-water foam
 - high-expansion foam



2. Carbon Dioxide Extinguishing Systems

- Carbon dioxide extinguishing systems are used for rooms that house electrical equipment, flammable liquid, and dry-cleaning machinery.

3. Dry-Chemical Piped Systems

- Dry-chemical piped systems used in situations where a quick extinguishing is needed, like a confined area.

4. Wet Chemical Extinguishing Systems

- Wet chemical systems are used to extinguish and control cooking and restaurant fires by dispersing a fine aerosol as well as a secondary saponification reaction with grease to prevent secondary auto-ignition fires.

5. Water Spraying & Automatic Sprinkler Systems

- Water-spray systems are effective with certain fires if no hazardous chemical reaction between water and the materials are burning. Used for extinguishing fires, exposure protection, and preventing fires by having water spray dissolve, dilute, disperse or cool flammable material.

6. Preventing Explosions

- Preventing explosions goes along with the fire protection in the design and construction of facilities.
- The maintenance and inspection of such facilities also ensures that qualified individuals are doing their best to prevent explosions.



7. Suppressing Explosions

- Explosion-suppression systems are designed to detect an explosion as it is starting.
- Once activated, the system will try to suppress, vent, or take other action to prevent the full explosive force.

